

OSI Model

Introduction

The Open Systems Interconnection (OSI) model is a seven-layer reference framework for network communication.

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Layer 7 - Application

Provides network services directly to end-user applications such as HTTP, FTP, SMTP and DNS.

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Layer 6 - Presentation

Handles data translation, encryption, compression and formatting.

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Layer 5 - Session

Establishes, manages and terminates communication sessions.

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Layer 4 - Transport

Provides reliable end-to-end delivery using TCP or fast delivery using UDP.

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Layer 3 - Network

Responsible for logical addressing and routing using IP.

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Layer 2 - Data Link

Handles framing, MAC addressing and error detection.

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Layer 1 - Physical

Transmits raw bits over cables, fiber or wireless media.

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OSI vs TCP/IP

OSI is a conceptual model while TCP/IP is the practical Internet protocol suite.

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Advantages

Standardization, modular design, interoperability and easier troubleshooting.

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Common Viva Questions

Explain the seven layers; Difference between OSI and TCP/IP; Functions of the Transport layer; Role of routers and switches.

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